Joshua Green

Electrical Engineer

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LinkedIn
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Education

Ph.D.
Electrical Engineering
Communication Systems and Data Networking
Network Security
Western University
Sept 2021 - Aug 2023
London, ON.

B.E.Sc
Electrical Engineering
Wireless Communication
Western University
Sept 2016 - Apr 2021
London, ON.

Skills

- **Allen Bradley PLC programming**
  - (FactoryTalk, DNET,RSLogix, RSLinx)
- **AutoCAD**
- **PCB Circuit Design**
  - (Microcap, EAGLE, Soldering)
- **MATLAB**
- **C++**
- **Python**
  - (Pandas, Keras, TensorFlow, Numpy, more)
- **Assembly**
- **Scripting**
  - (Linux, Bash)
- **Git**
- **HTML/CSS**
- **Engineering tools**
  - (QuartusII, Arduino, SOLIDWORKS)
- **Project Management**
- **Welding**
  - (MIG, Stick, Oxy-Acetylene)
- **Manufacturing**
  - (Lathe, Machining, Plasma-cutting)

Overview

I am a highly motivated Electrical Engineering Graduate with a diversity of interests and expertise. My multi-domain knowledge from other interests and experience in my life put me ahead of everyone else around me. My curiosity, ability to learn quickly along with my tenacious passion put towards my work paired with the ability to critically think and easily apply what I learn throughout my schooling and career sets me apart from the herd.

Publications and Awards

- IEEE PIMRC Conference Paper Publication:
  "Shared Experiential History for Encryption Based Peer-To-Peer Authentication"
- 2023 Graduate Symposium Outstanding Presentation in Communication Systems Engineering
- Dean's List Honoree 2017-2018

Projects

- Lung disease classification using CNN and Transfer learning
- Capstone: Smart cane elderly alert system
  - Capstone role: GPS data polling, data formatting, SMS message formatting, send SMS to contacts
- Project 2: Welder Power Supply with PID
- Project 1: Student card automatic door access

Work Experience

Graduate Research and Teaching Assistant
Western University - London, ON Sept 2021 - Current

- Drive pioneering research in wireless network security.
- Engineer novel methods for Authentication and Trust Assessment
- Monitor and mark labs, midterm and final exams, and assignments for the following courses:
  - ECE3331B (Introduction to Signal Processing)
    - Course is heavily reliant on MATLAB and Simulink.
  - ECE4436A (Networking: Principles, Protocols and Architectures)
    - Linux · Socket Programming · Virtual Machines · Python
  - ECE2236B (Magnetic Circuits and Transmission Lines)
    - Electromagentic Theory

Electrical Engineering Intern (Controls)
General Motors - Ingersoll, ON May 2019 - May 2020

- Create original PLC program solutions using Allen Bradley PLC Software (Factory Talk, DNET, RSLogix, RSLinx),
- Departmental DNET backup during the plant shutdown down due to first covid-19 wave,
- Update electrical drawings in AutoCAD and document the changes in the company database,
- Cross-functional teamwork at a large scale,
- Quality defect isolation and solutions,
- Project Management (PM delegation, SAP ordering, supervising engineering projects).
- Upholding Ontario Electrical Safety Code (OESC).
- Update company database for: Lockout Zones/Tags, Safety zones, PLC programs, DNET.